A Capability-Based Comparative Review of Urban Digital Twin Architectures for Real-Time Monitoring and Decision Support

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Abstract—This paper presents a capability-based comparative review of 20 peer-reviewed research studies on urban Digital Twins (DTs) for smart-city monitoring and decision support. Rather than proposing another generic DT lifecycle, the review evaluates the selected studies using a common capability framework covering data sourcing, real-time updating, data integration, artificial intelligence, prediction, simulation, visualization, decision support, actuation, scalable computing, and security or governance. A structured review protocol was used to define the corpus, exclude survey/review papers from the 20-study core set, and classify each study by publication type and demonstrated capabilities. The resulting matrix enables quantitative comparison of capability coverage across the literature. The analysis shows that multi-source data handling, integration, and decision-oriented analytics are common, while physical actuation and explicit security or governance treatment are much less frequently demonstrated. Mobility and traffic studies provide the strongest evidence for operational real-time twins, whereas environmental and resilience studies emphasize spatial modeling and scenario analysis. The review therefore contributes a reproducible comparison structure and an evidence-based synthesis of where current urban DT research is concentrated and where integrated end-to-end capability remains limited.

Keywords—*digital twin, smart city, urban digital twin, comparative review, capability matrix, real-time monitoring, decision support, predictive analytics*

I. INTRODUCTION

Urban Digital Twins (UDTs) are increasingly used to represent physical city systems through connected digital models, live or periodically updated data, analytics, simulation, and visualization. In smart-city settings, the physical scope can include transport networks, buildings, utilities, environmental sensors, public infrastructure, vehicles, and citizen-generated information. The practical value of a UDT therefore depends not only on the existence of a virtual representation, but on the capabilities that connect observation, integration, analysis, prediction, decision support, and—when demonstrated—physical intervention.

Recent research illustrates substantial variation in how these capabilities are implemented. Mendula et al. [1] describe a data-driven urban activity twin with descriptive and predictive functions, while Kim and Ham [2] use participatory sensing and deep learning to update representations of built infrastructure. Mobility studies move further toward operational use: Xu et al. [3] demonstrate real-time situational awareness and cyber-physical traffic control, and Bellini et al. [4] present a scalable platform for mobility decision support using microservices and AI/ML/RL. Environmental and resilience studies emphasize different requirements, such as spatial accuracy, model–sensor integration, and scenario analysis [5], [6], [7].

At the same time, recent review literature has already covered broad smart-city DT domains and implementation challenges. A 2023 review by Therias and Rafiee focuses on how city digital twins can support urban resilience and combines literature analysis with expert interviews [8]. A 2025 comprehensive review categorizes enabling technologies and applications across multiple smart-city domains [9]. Most directly relevant to the present study, Yessef et al. [10] provide an implementation-oriented review that organizes capabilities into an operational pipeline and introduces a capability/use-case matrix and implementation-readiness scale. These contributions establish that another broad narrative review, or another generic lifecycle, would add limited value.

The specific gap addressed here is therefore comparative rather than definitional. The existing review literature provides broad domain coverage, resilience-focused interpretation, or implementation-oriented lifecycle guidance, but the 20 selected non-review research studies in the present corpus have not been evaluated within one consistent capability coding scheme that makes capability coverage directly comparable across studies. The objective of this review is to fill that gap by constructing a capability-based comparison of the 20 studies and quantifying the presence of key capabilities. The review asks three questions: (RQ1)

Which technical and operational capabilities are demonstrated across the selected studies? (RQ2) How frequently does each capability occur, and which combinations remain uncommon? (RQ3) What research gaps become visible when the studies are compared using the same criteria?

The contribution is intentionally limited to a review contribution. This paper does not claim to invent a new Digital Twin architecture or a new city-control pipeline. Instead, it contributes a reproducible comparison framework, a coded capability matrix, a quantitative synthesis of the selected evidence, and an evidence-derived interpretation of the remaining integration gaps.

II. REVIEW METHODOLOGY

A. *Review Scope and Corpus*

The review corpus contains 20 peer-reviewed research studies addressing urban or smart-city Digital Twin applications, architectures, data integration, simulation, or decision support. The core set excludes survey papers, systematic reviews, bibliometric studies, and review-only articles. The publication years represented in the final corpus are 2021–2026; the inclusion of the 2021 disaster-decision study [7] is deliberate and is reported explicitly rather than being described as part of a 2022–2026-only set.

B. *Source Classification*

The source mix was also recorded accurately. Sixteen of the 20 selected studies are IEEE publications; the remaining four are published through other scholarly venues: the Journal of Computational Design and Engineering [11], the ASCE Construction Research Congress proceedings [2], Procedia Computer Science [12], and Nature Computational Science [7]. Thus, the corpus is described as 20 peer-reviewed research studies, not as 20 IEEE papers.

C. *Search Strategy*

The revised protocol uses the following core Boolean expression for discovery: ("digital twin" OR "urban digital twin" OR "city digital twin") AND ("smart city" OR urban) AND ("real-time monitoring" OR synchronization OR prediction OR simulation OR "decision support" OR mobility OR traffic OR environment). Application-specific terms such as "air quality," flood, water, building, resilience, and participatory sensing were used as secondary searches when needed to identify directly relevant studies. Discovery sources were IEEE Xplore, Scopus, Web of Science, ScienceDirect, and SpringerLink; Crossref and publisher records were used for bibliographic verification. The search window was 2021–2026 so that the final corpus remains consistent with the oldest included study.

D. *Inclusion and Exclusion Criteria*

Studies were included when they (i) addressed a Digital Twin or city-scale digital-twin application, (ii) concerned a smart-city or urban-management problem, (iii) reported a research contribution rather than only a conceptual review, and (iv) provided enough information to identify its data sources, methods, architecture, application, or decision function. Studies were excluded when they were surveys or systematic reviews, unrelated industrial/product twins, purely theoretical papers without a relevant urban contribution, duplicates, or papers whose primary focus was outside smart-city management.

E. *Screening and Coding Procedure*

Screening was performed in three stages: title/abstract relevance, full-text or publisher-record relevance, and final eligibility against the inclusion criteria. Duplicates and review-only records were removed before the 20-study core set was fixed. Each included study was then coded against the same capability definitions. A capability was marked present only when it was explicitly demonstrated or described as part of the study contribution; a capability was not inferred merely because the technology could theoretically support it. This rule is important for distinguishing a reported capability from a possible future extension.

F. *Capability Definitions*

The comparison framework uses eleven dimensions: multi-source data (DS), real-time/update capability (RT), integration or semantic interoperability (INT), AI/ML (AI), prediction/forecasting (PR), simulation or scenario analysis (SIM), visualization or 3D/AR/MR (VIS), decision support (DEC), physical actuation/control (ACT), scalable or distributed computing (SC), and explicit security/governance treatment (SEC). The first ten dimensions are used for the main quantitative comparison; SEC is retained as a cross-cutting gap indicator because explicit security/governance treatment is not consistently reported in the corpus.

III. CAPABILITY-BASED COMPARATIVE FRAMEWORK

The framework is designed specifically to compare studies rather than to propose a new Digital Twin lifecycle. It treats a UDT as a set of observable capabilities. A study may therefore score positively for data integration and prediction without necessarily providing physical actuation, while another may emphasize visualization or semantic modeling without real-time operation. This avoids forcing every paper into one architecture and makes differences between application-oriented and platform-oriented studies visible.

DS captures whether the study combines more than one meaningful urban data source, such as IoT sensors, GIS, traffic data, citizen information, administrative records, or environmental measurements. RT captures explicit live, online, continuously updated, or real-time processing. INT captures data integration, semantic modeling, common urban entities, or integration of heterogeneous data streams. AI captures machine learning, artificial intelligence, reinforcement learning, or related data-driven intelligence. PR captures forecasting or prediction of future states or indicators. SIM captures explicit simulation, what-if analysis, geosimulation, or scenario evaluation. VIS captures 3D city models, visualization, augmented reality, mixed reality, or related human-facing representations. DEC captures explicit decision support, planning, recommendation, optimization, or intervention selection. ACT is reserved for demonstrated physical control or cyber-physical actuation rather than a generic statement that the twin could control something. SC captures edge-cloud, microservices, cloud, distributed, or other explicit mechanisms for handling scale. SEC records

explicit security, privacy, identity, provenance, or governance mechanisms.

The framework differs from existing reviews in both unit of analysis and output. The 2023 resilience review [8] organizes findings around resilience concepts, stakeholder needs, and technical implementation. The 2025 comprehensive review [9] compares enabling technologies and smart-city domains. Yessef et al. [10] organize capabilities into an implementation-oriented pipeline and readiness scale. The present review instead treats the individual research study as the coding unit and produces a capability-by-study matrix followed by numerical coverage statistics. Consequently, the framework is not intended to replace those reviews; it provides a narrower comparative lens over a fixed 20-study research corpus.

Because established Digital Twin pipelines already appear in the literature, the earlier sequence "sense, integrate, synchronize, model, predict, simulate, decide, act, and learn" is not retained as an original architectural proposal. Where useful, related stages are discussed only as an evidence-derived interpretation of the capability patterns observed in the matrix.

IV. COMPARATIVE CAPABILITY MATRIX

The matrix below is the principal comparison instrument of this review. DS = multi-source data; RT = real-time/update; INT = integration/semantics; AI = artificial intelligence or machine learning; PR = prediction; SIM = simulation/scenario analysis; VIS = visualization/3D/AR/MR; DEC = decision support; ACT = physical actuation/control; SC = scalable/distributed computing. A check mark means explicit evidence in the selected study; a dash means that the capability was not identified under the coding rule.

DS=data sources; RT=real-time/update; INT=integration; AI=AI/ML; PR=prediction; SIM=simulation; VIS=visualization; DEC=decision support; ACT=actuation; SC=scalable/distributed.

Study ID [Ref.]	DS	RT	AI	PR	DEC
1 [1]	✓	✓	✓	✓	✓
2 [18]	✓	✓	–	✓	✓
3 [5]	✓	–	–	–	–
4 [11]	✓	–	–	–	–
5 [2]	✓	✓	✓	–	✓
6 [19]	✓	–	✓	–	✓
7 [6]	✓	–	✓	–	✓
8 [3]	✓	✓	✓	✓	✓
9 [20]	✓	–	✓	–	✓
10 [12]	✓	–	–	–	✓

Table I-A. Core capability coding for Studies 1–10 (✓ = explicit evidence; – = not identified).

Study ID [Ref.]	DS	RT	AI	PR	DEC
11 [17]	✓	✓	✓	✓	✓
12 [16]	✓	–	–	–	–
13 [21]	✓	✓	✓	✓	✓
14 [13]	✓	✓	✓	✓	✓
15 [22]	✓	–	–	–	✓
16 [14]	✓	✓	✓	✓	✓
17 [4]	✓	✓	✓	✓	✓
18 [23]	✓	✓	–	✓	✓
19 [15]	✓	✓	✓	✓	✓
20 [7]	–	–	–	–	✓

Table I-B. Core capability coding for Studies 11–20 (✓ = explicit evidence; – = not identified).

Complementary dimensions coded in the full matrix are INT (integration/semantics), SIM (simulation/scenario), VIS (visualization), ACT (physical actuation/control), and SC (scalable/distributed computing). Their corpus-level counts are reported in Table II.

IV. COMPARATIVE ANALYSIS OF THE 20 STUDIES

The capability coding reveals a heterogeneous research landscape. Table I provides the study-level mapping used with the capability matrix. A check mark in the matrix indicates that the capability is explicitly demonstrated or described in the selected study; a dash indicates that the capability was not identified under the coding rule. The matrix is deliberately more restrictive than a general literature summary: possible capabilities are not counted unless they are supported by the study.

Studies [1], [2], [3], [13], [14], [4], and [15] show comparatively strong combinations of live data, analytics, and urban-management functions. Mobility research is especially prominent because traffic systems naturally provide frequent data streams and measurable operational outcomes. In contrast, [11] and [16] concentrate on representation and semantics, respectively, illustrating that not every UDT contribution is intended to be a real-time control system.

Environmental and resilience studies demonstrate another pattern. Air-quality work [5], [12] and water/flood work [6] emphasize combining measurements with spatial or mathematical models. Disaster decision support [7] emphasizes scenario reasoning under uncertainty. These studies strengthen the simulation and spatial dimensions of UDT research but do not necessarily provide continuous physical control.

Architecture studies [17], [13], [14], and [4] concentrate on reusable platforms, integration, distributed processing, or scalable services. These papers are important for city-scale deployment because a UDT must handle heterogeneous sources and large data volumes. However, platform scalability and decision intelligence are not equivalent to closed-loop actuation. The matrix therefore keeps those capabilities separate.

V. QUANTITATIVE COMPARISON

Using the final capability matrix, the most common capabilities are multi-source data and integration/semantic handling. Decision support is also frequent, reflecting the shift from static visualization toward planning, optimization, and operational use. AI/ML appears in a substantial subset of the studies, while prediction and simulation are present in roughly half of the corpus. Visualization is similarly common, reflecting the continued importance of 3D and immersive representations.

The least common operational capability is physical actuation/control. Only a small number of studies in the selected corpus explicitly demonstrate a link from the twin to physical control; the clearest example is the traffic-control work of Xu et al. [3]. Security/governance is even less consistently reported as a primary capability in the selected research studies. This does not mean security is unimportant; rather, it indicates that security, privacy, provenance, and governance are frequently treated as requirements or future

work instead of being integrated into the demonstrated technical contribution.

The quantitative pattern also exposes an important distinction between analytical intelligence and operational closure. Prediction and simulation can be well developed while actuation remains absent. Similarly, a platform can integrate many sources and provide visualization without demonstrating validated intervention. The comparative framework therefore prevents a high count of general capabilities from being interpreted as evidence of a fully closed-loop city twin.

The results should be interpreted as descriptive statistics of this selected 20-study corpus, not as prevalence estimates for the entire Digital Twin field. The corpus is purposive and application-focused, and its composition contains a strong mobility and architecture component. The value of the counts is therefore comparative: they reveal the capability distribution inside the reviewed set and provide a transparent basis for identifying gaps.

Capability	Coverage of 20 studies
DS Multi-source data	19/20 (95%)
RT Real-time/update	11/20 (55%)
INT Integration/semantics	19/20 (95%)
AI/ML	12/20 (60%)
PR Prediction	10/20 (50%)
SIM Simulation/scenario	9/20 (45%)
VIS Visualization/3D	10/20 (50%)
DEC Decision support	17/20 (85%)
ACT Physical actuation/control	2/20 (10%)
SC Scalable/distributed	10/20 (50%)

Table II. Quantitative capability coverage in the selected corpus.
Note: These are descriptive statistics of the selected 20-study corpus, not prevalence estimates for the entire urban Digital Twin literature.

VI. HOW THIS REVIEW DIFFERS FROM EXISTING REVIEWS

Existing reviews establish important foundations but use different analytical units and purposes. Therias and Rafiee [8] examine city Digital Twins through the lens of urban resilience and use literature plus expert interviews to explore understanding, future imagination, collaboration, and implementation requirements. Their focus is resilience-oriented rather than a capability-by-study quantitative comparison.

The 2025 comprehensive review [9] provides broad coverage of smart-city domains and enabling technologies, including IoT, machine learning, cyber-physical systems, blockchain, transportation, water, energy, public safety, and other domains. Its contribution is breadth across the smart-city landscape. The present review narrows the scope to 20 selected research studies and applies one common capability code to each study.

Yessef et al. [10] provide the closest methodological comparison. Their review introduces an implementation-oriented capability pipeline, measurable engineering targets, a capability/use-case matrix, and a Digital-Twin Implementation Readiness Level. Those elements are valuable established contributions. The present paper does not reproduce them as original ideas. Instead, it uses a study-level capability matrix to compare the selected research papers and quantify which capabilities are actually demonstrated in the corpus.

The resulting distinction is therefore threefold: (1) the unit of analysis is the individual research study; (2) the primary output is a transparent capability-by-study coding matrix and numerical coverage summary; and (3) the research gap is expressed as missing capability combinations and under-demonstrated operational functions rather than as another generic Digital Twin lifecycle. This positioning gives the review a specific comparative contribution while respecting prior frameworks.

VII. RESEARCH GAPS AND EVIDENCE-DERIVED SYNTHESIS

Four gaps become particularly clear from the comparison. First, interoperability is still uneven. Studies use different data models, domain representations, IoT platforms, and application-specific interfaces. Semantic work such as [16] addresses part of this problem, but semantic consistency is not uniformly integrated into real-time operational platforms. A future UDT should therefore treat shared urban entities and machine-readable semantics as core integration mechanisms rather than optional documentation.

Second, real-time capability does not automatically imply reliable synchronization. Live data can be delayed, missing, noisy, or spatially incomplete. The air-quality study [5] shows why model–sensor coupling and calibration matter, while participatory sensing [2] illustrates dependence on image quality and citizen participation. Real-time UDT research should therefore evaluate not only update frequency but also data quality, uncertainty, synchronization error, and the effect of imperfect observations on downstream decisions.

Third, the literature has a noticeable gap between prediction/simulation and validated physical intervention. Traffic research [3] provides evidence of cyber-physical control, but many other studies stop at monitoring, visualization, forecasting, or scenario evaluation. This suggests a need for safe human-in-the-loop control, intervention validation, feedback measurement, and explicit criteria for when an automated recommendation is trusted.

Fourth, security, privacy, governance, and accountability are not consistently demonstrated as integrated capabilities. City twins can combine traffic, environmental, building, mobility, and citizen information, creating technical and institutional risks. Future work should connect identity and access management, privacy-preserving analytics, provenance, auditability, model governance, and uncertainty communication with the same operational workflows used for analytics and decision support.

The evidence-derived synthesis can therefore be stated as a progression of capability dependencies rather than a new architecture: reliable urban data must be collected and integrated before a twin can be updated; synchronized data supports state estimation; validated models support prediction and simulation; these outputs can support human or automated decisions; and only selected studies currently demonstrate a validated transition from decisions to physical control. Feedback from interventions is then required to evaluate whether the predicted effect actually occurred. This synthesis is deliberately described as an interpretation of the

comparative evidence, not as an original Digital Twin pipeline.

VIII. IMPLICATIONS OF THE COMPARATIVE FINDINGS

The comparison also suggests that future UDT studies should be evaluated in terms of capability combinations rather than isolated functions. Prediction without reliable data integration can produce unstable results, while visualization without decision support may remain primarily descriptive. A useful research evaluation should therefore report the chain of evidence connecting source data, synchronization, model execution, analytical output, decision recommendation, and observed outcome. This makes it possible to distinguish a demonstrator that displays a city model from a system that supports validated operational decisions.

A second implication concerns evaluation standards. The selected studies use different application metrics, making direct technical comparison difficult. Future work would benefit from reporting common measures such as update latency, data completeness, synchronization error, prediction accuracy, simulation execution time, decision-response time, resource consumption, and control reliability. Reporting these measures alongside the capability matrix would allow future reviews to compare not only whether a capability exists, but how effectively it is implemented.

A third implication is the need for benchmarkable urban datasets and repeatable scenarios. Traffic, air-quality, flood, building, and public-infrastructure twins are often evaluated in different cities with different sensor configurations. Shared benchmark scenarios could allow researchers to test interoperability, model transfer, uncertainty handling, and scalability under comparable conditions. Such benchmarks would also make it easier to assess whether an architecture that performs well in one demonstrator can be adapted to another urban context.

The framework also has a practical implication for municipal planning. City authorities may not need every capability at the same time. A monitoring-focused deployment may prioritize reliable data integration and visualization, while a traffic-control deployment may require low-latency synchronization, prediction, decision support, and validated actuation. The matrix can therefore be used as a requirements checklist when a city defines the scope of a proposed twin. This is a review-derived use of the framework, not a prescriptive architecture for every city.

Finally, deployment maturity should be treated separately from feature count. A platform with many listed technologies is not necessarily closer to operational deployment than a smaller system with reliable synchronization and validated control. Future comparative studies should report the maturity of the demonstrated workflow, the role of human operators, the quality of feedback data, and the safeguards applied before physical intervention. This would complement capability-based comparison without claiming that a particular maturity scale is newly invented by this review.

IX. CONCLUSION

This review compared 20 peer-reviewed urban Digital Twin research studies using a common capability-based framework. The approach addresses a specific limitation of broad narrative reviews: individual studies often use different terminology and emphasize different functions, making direct comparison difficult. By coding the same capability dimensions across the corpus and summarizing their occurrence quantitatively, the review identifies where the evidence is concentrated and where important functions remain under-demonstrated.

The analysis indicates that multi-source data, integration, decision support, and data-driven intelligence are central themes. Prediction, simulation, visualization, and scalable computing are also substantial components of the selected research. In contrast, explicit physical actuation and integrated security/governance are uncommon. Mobility studies provide some of the clearest examples of real-time operational behavior, while environmental, resilience, semantic, and building studies contribute complementary capabilities.

The paper's contribution is therefore a comparative review method and evidence synthesis, not a claim of a new urban Digital Twin architecture. The capability matrix can be reused for future studies and expanded as additional city-scale Digital Twin implementations become available. The next research step should be to evaluate these capabilities using common datasets, performance measures, uncertainty reporting, and intervention-validation protocols so that urban Digital Twins can progress from isolated demonstrations toward trustworthy decision-support infrastructure.

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